

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286180

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Han; Won Chull et al.

SECONDARY BATTERY

Abstract

A secondary battery may include a can having one end opened, an electrode assembly accommodated in the can, a cap plate coupled to the one end of the can that is opened, and a gasket surrounding at least a portion of top and bottom surfaces of the cap plate and integrated with the cap plate. According to some embodiments of the present disclosure, the cap plate and the gasket may be integrated with each other to reduce costs and assembly time. In addition, the thickness of the gasket at the lower portion of the cap plate may be reduced to secure the space for improving the battery capacity while maintaining the sealing performance.

Inventors: Han; Won Chull (Yongin-si, KR), Kim; Young Ho (Yongin-si, KR)

Applicant: Samsung SDI Co., Ltd. (Yongin-si, KR)

Family ID: 96949524

Assignee: Samsung SDI Co., Ltd. (Yongin-si, KR)

Appl. No.: 18/794872

Filed: August 05, 2024

Foreign Application Priority Data

KR 10-2024-0033612

Mar. 11, 2024

Publication Classification

Int. Cl.: H01M50/186 (20210101); H01M50/107 (20210101); H01M50/152 (20210101);
H01M50/184 (20210101); H01M50/198 (20210101); H01M50/533 (20210101);
H01M50/586 (20210101); H01M50/59 (20210101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This present application claims priority to and the benefit under 35 U.S.C. § 119(a)-(d) of Korean Patent Application No. 10-2024-0033612, filed on Mar. 11, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

FIELD

[0002] Embodiments relate to a secondary battery.

BACKGROUND

[0003] A cylindrical secondary battery may include a cylindrical can having one end opened and having a cap plate coupled to said one end that is opened. In general, the cap plate may be insulatingly coupled to a can (e.g., coupled through a gasket or insulator made of insulating material). The gasket can function to insulate the can from the cap plate and seal the can at the same time.

[0004] The above-described information disclosed in the technology that serves as the background of the present disclosure is only for improving understanding of the background of the present disclosure and thus may include information that does not constitute the related art.

SUMMARY

[0005] Aspects of some embodiments of the present disclosure provide a secondary battery having an improved structure of a cap assembly.

[0006] According to some embodiments, a secondary battery includes: a can having one end opened; an electrode assembly accommodated in the can; a cap plate coupled to the one end of the can; and a gasket configured to surround at least a portion of top and bottom surfaces of the cap plate and integrated with the cap plate.

[0007] The can may have a cylindrical shape, and the cap plate may have a circular plate shape.

[0008] The gasket may have a ring shape with a predetermined width and include a first portion disposed on the top surface of the cap plate, a third portion disposed on the bottom surface of the cap plate, and a second portion configured to connect the first portion to the third portion.

[0009] The cap plate may include a coupling part disposed on an area of the top surface that is in contact with the first portion.

[0010] The coupling part may have a sawtooth shape in cross-section.

[0011] The coupling part may be disposed in a circumferential direction of the cap plate.

[0012] The coupling part may be manufactured through a knurling method.

[0013] A height of the coupling part may be within about 30% of a thickness of the cap plate.

[0014] The gasket and the cap plate may be provided through double injection.

[0015] The gasket may be made of an insulating material.

[0016] In the gasket, a thickness of the third portion may be less than that of the first portion.

[0017] The cap plate may include a notch disposed along a circumferential direction.

[0018] The cap plate may include a first area, a second area disposed inside the first area and provided to be concave in a direction of the electrode assembly, and a third area disposed outside the first area and disposed in the same line as the second area.

[0019] The notch may be provided in the first area.

[0020] The gasket may be configured to surround the third area and may be spaced apart from the notch.

[0021] The gasket may further include an extension portion protruding from an end of the third portion toward the electrode assembly.

[0022] The secondary battery may further include a first current collector plate electrically connected to a positive electrode plate of the electrode assembly; and a second current collector plate electrically connected to a negative electrode plate of the electrode assembly, wherein the extension portion may have a length at which the extension portion is not in contact with the second current collector plate.

[0023] The can may include a circular top surface part and a side part extending downward from the top surface part, and the second current collector plate may include a plurality of negative electrode leads electrically connected to the side part.

[0024] The can may further include: a beading part provided to be concave inward from the side part; and a crimping part spaced apart from the beading part and provided by bending an end of the side part inward, wherein a negative electrode lead of the plurality of negative electrode leads may be inserted between the beading part and the second gasket.

[0025] The secondary battery may further include: a positive electrode terminal inserted into the top surface part and electrically connected to the first current collector plate; and a gasket disposed between the top surface part and the positive electrode terminal and made of an insulating material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. 1 illustrates a perspective view of a secondary battery, according to some embodiments;

[0027] FIG. 2 illustrates a cross-sectional view of the secondary battery of FIG. 1;

[0028] FIG. 3 illustrates a partial perspective view of an arrangement of a cap assembly and an electrode assembly of the secondary battery of FIG. 2;

[0029] FIG. 4 illustrates a front view of a portion of the cap assembly and a portion of the electrode assembly of the secondary battery of FIG. 2;

[0030] FIG. 5 illustrates a perspective view of the cap assembly of the secondary battery of FIG. 2;

[0031] FIG. 6 illustrates a rear perspective view of the cap assembly of the secondary battery of FIG. 2;

[0032] FIG. 7A illustrates a partial cross-sectional view of a portion of the cap assembly, taken along line A-A in FIG. 5; and

[0033] FIGS. 7B illustrates a partial cross-sectional view of the cap assembly of FIG. 7A, according to other embodiments.

DETAILED DESCRIPTION

[0034] The present disclosure may be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided so that those skilled in the art thoroughly understand the present disclosure. Further, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art.

[0035] In addition, in the accompanying drawings, the thickness or size of each layer is exaggerated for simplicity and clarity of description, and the same reference numerals in the drawings refer to the same elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. In this specification, it will also be understood that if a member A is referred to as being connected to a member B, the member A may be directly connected to the member B or indirectly connected to the member B with a member B therebetween.

[0036] The terms used herein are for illustrative purposes of the present disclosure only and should

not be construed to limit the meaning or the scope of the present disclosure. As used in this specification, a singular form may, unless definitely indicating a particular case in terms of the context, include a plural form. Also, the expressions “comprise/include” and/or “comprising/including” used in this specification specify the presence of but neither define the mentioned shapes, numbers, steps, operations, members, elements, and/or groups of these, nor exclude the presence or addition of one or more other different shapes, numbers, steps, operations, members, elements, and/or groups of these, or addition of these. The term “and/or” used herein includes any and all combinations of one or more of the associated listed items.

[0037] As used herein, terms such as “first,” “second,” etc. are used to describe various members, components, areas, layers, and/or portions. However, it is obvious that the members, components, areas, layers, and/or portions should not be defined by these terms. The terms do not mean a particular order, up and down, or superiority, and are used only for distinguishing one member, component, region, layer, or portion from another member, component, region, layer, or portion. Thus, a first member, component, region, layer, or portion which will be described may also refer to a second member, component, region, layer, or portion, without departing from the teaching of the present disclosure.

[0038] Spatially relative terms, such as “below”, “beneath”, “lower”, “above”, “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as shown in the figures. These spatially relative terms are intended for ease of comprehension of the present disclosure according to various process states or usage states of the present disclosure, and thus, the present disclosure is not limited thereto. For example, if an element or feature shown in the drawings is turned inside out, the element or feature described as “beneath” or “below” may change into “above” or “upper”. Thus, the term “lower” may encompass the term “upper” or “below”.

[0039] The term “about” may be used to mean within $\pm 20\%$ of a target value in some embodiments, within $\pm 10\%$ of a target value in some embodiments, and within $\pm 5\%$ of a target value in some embodiments. The term “about” may equal the target value.

[0040] Hereinafter, a secondary battery, according to some embodiments of the present disclosure, will be described in detail with reference to the accompanying drawings.

[0041] As described herein, a cylindrical secondary battery may include a cylindrical can having one end opened and having a cap plate coupled to said end. In general, the cap plate may be coupled through a gasket or insulator made of insulating material. The gasket can function to insulate the can from the cap plate and seal the can.

[0042] As a thickness of the gasket increases, sealing performance is improved, but as the thickness of the gasket increases, a size of the electrode assembly has to be reduced, which reduces a capacity of the secondary battery. Thus, it is desirable to provide a structure that is capable of increasing the capacity of the secondary battery while maintaining the sealing function.

[0043] FIG. 1 illustrates a perspective view of a secondary battery, according to some embodiments. FIG. 2 illustrates a cross-sectional view of the secondary battery of FIG. 1. FIG. 3 illustrates a partial perspective view of an arrangement of a cap assembly and an electrode assembly of the secondary battery of FIG. 2. FIG. 4 illustrates a front view of a portion of the cap assembly and a portion of the electrode assembly of the secondary battery of FIG. 2. FIG. 5 illustrates a perspective view of the cap assembly of the secondary battery of FIG. 2. FIG. 6 illustrates a rear perspective view of the cap assembly of the secondary battery of FIG. 2. FIG. 7A illustrates a partial cross-sectional view of a portion of the cap assembly, taken along line A-A in FIG. 5.

[0044] Referring to FIG. 1, a secondary battery, according to embodiments, may be a cylindrical battery having a cylindrical outer appearance.

[0045] Referring to FIGS. 1 to 4, a secondary battery 10, according to some embodiments, may include cap assemblies (e.g., including cap plate 800 and/or second gasket 900). The secondary

battery **10** includes a can **100**, an electrode assembly **200**, a first current collector plate **300**, a second current collector plate **400**, an insulating member **500**, a positive electrode terminal **600**, a first gasket **700**, a cap plate **800**, and a second gasket **900**.

[0046] Referring to FIGS. **1** and **2**, the can **100** may accommodate the electrode assembly **200** and an electrolyte and may have a substantially cylindrical shape. The can **100** may include a circular top surface part **110** and a side part **120** extending downward from the top surface part **110**. The positive electrode terminal **600** and the first gasket **700** may be coupled to the top surface part **110**. In some embodiments, a terminal hole (not shown in FIG. **1**) may be defined to pass through a center of the top surface **110**. A beading part **122** and a crimping part **124** may be provided at a lower portion of the side part **120**. In some embodiments, an example secondary battery in which a lower portion of the can **100** is opened is described, but conversely, an upper portion of the can **100** may be opened.

[0047] In some embodiments, when assembling the electrode assembly **200** into the can **100**, the top surface part **110** may be disposed downward, the electrode assembly **200** may be inserted, and the beading part **122** may be provided. The beading part **122** may prevent the electrode assembly **200** from being separated. The beading part **122** may be provided by processing a lower end of the side part **120** so as to be concave toward the inside of the can **100** (e.g., the beading part may be provided to be concave inward from the side part **120**). After providing the beading part **122**, the cap plate **800** and the second gasket **900** may be assembled, and the crimping part **124** may be provided to prevent the cap plate **800** from being separated. The crimping part **124** may be spaced apart from the beading part and may be provided by bending an end of the side part **120** toward the inside of the can **100**.

[0048] Referring to FIGS. **2** to **4**, the electrode assembly **200** may include a first electrode plate **210**, a second electrode plate **220**, and a separator **230**.

[0049] The first electrode plate **210** may be any one of a negative electrode plate and a positive electrode plate. In some embodiments, the description will be based on an example in which the first electrode plate **210** is a positive electrode plate. The first electrode plate **210**, which may be the positive electrode plate, may be provided as a thin metal plate having excellent conductivity, for example, aluminum (Al) foil or mesh. The first electrode plate **210** may be provided with a positive electrode coated portion coated with a positive electrode active material and a positive electrode non-coated portion that is not coated with the positive electrode active material. For example, the positive electrode active material may be formed of a chalcogenide compound, for example, composite metal oxide such as $\text{LiCoO}_{0.5}\text{Li}_{0.5}\text{O}$, $\text{LiMn}_{0.5}\text{Li}_{0.5}\text{O}_{2.5}$, $\text{LiNiO}_{0.5}\text{Li}_{0.5}\text{O}$, $\text{LiNiMnO}_{0.5}\text{Li}_{0.5}\text{O}$, and the like.

[0050] The second electrode plate **220** may be the other of the negative electrode plate and the positive electrode plate. In some embodiments, the description will be based on an example in which the second electrode plate **220** is a negative electrode plate. The second electrode plate **220**, which may be the negative electrode plate, may be provided as a conductive metal thin plate, for example, copper (Cu) or nickel (Ni) foil or mesh. The second electrode plate **220** may be provided with a negative electrode coated portion coated with a negative electrode active material and a negative electrode non-coated portion that is not coated with the negative electrode active material. For example, the negative electrode active material may include a carbon-based material, silicon (Si), tin (Sn), tin oxide, a tin alloy composite, transition metal oxide, lithium metal nitrite, or metal oxide.

[0051] The separator **230** may be interposed between the first electrode plate **210** and the second electrode plate **220** such as to serve to prevent short circuit between the first electrode plate **210** and the second electrode plate **220**. In some embodiments, the separator **230** may be made of polyethylene, polypropylene, a porous copolymer of polyethylene and polypropylene, or the like.

[0052] For example, the electrode assembly **200** may be disposed so that the non-coating portion of the first electrode plate **210** protrudes upward from an upper end of the second electrode plate **220**.

In some embodiments, the non-coating portion of the second electrode plate **220** may be disposed to protrude downward from a lower end of the first electrode plate **210** and may be wound in a jelly-roll shape. In this state, the first current collector plate **300** may be welded to the non-coating portion of the first electrode plate **210**, and the second current collector plate **400** may be welded to the non-coating portion of the second electrode plate **220**.

[0053] Referring to FIG. 2, the first current collector plate **300** may be disposed between the top surface part **110** and the non-coated portion of the first electrode plate **210**. The first current collector plate **300** may be welded and electrically connected to the first electrode plate **210**, which may be the positive electrode plate, and thus may be defined as a positive current collector plate. The first current collector plate **300** may have an approximately circular plate shape. The first current collector plate **300** may be electrically connected to the positive electrode terminal **600** at a central portion thereof. However, the first current collector plate **300** has to be insulated from the can **100**. In some embodiments, the insulating member **500**, which will be described later, may be provided between the first current collector plate **300** and the can **100**.

[0054] Referring to FIGS. 2 to 4, the second current collector plate **400** may be disposed between the cap plate **800** and the non-coated portion of the second electrode plate **220**. The second current collector plate **400** may be welded and electrically connected to the second electrode plate **220**, which may be the negative electrode plate, and thus may be defined as a negative electrode current collector plate. The second current collector plate **400** may have an approximately circular plate shape. A plurality of negative electrode leads **410** may protrude from the second current collector plate **400** (e.g., the second current collector plate may include a plurality of leads). The second current collector plate **400** may be electrically connected to the side part **120** of the can **100** through the negative electrode leads **410** (e.g., the negative electrode leads electrically connected to the side part).

[0055] Each of the negative electrode leads **410** may be in the form of a plate having a predetermined length and width. The negative electrode lead **410** may be in contact with the beading part **122** to be welded. The negative electrode lead **410** may be fixed by the beading part **122**, the second gasket **900**, and the crimping part **124**, which will be described later, without welding. The negative electrode lead may be inserted between the beading part and the second gasket. Therefore, the welding between the negative electrode lead **410** and the beading part **122** may be omitted. In some embodiments, the second current collector plate **400** may be directly welded to the side part **120** and electrically connected without the negative electrode lead **410**. Because the negative electrode lead **410** is electrically connected to the side part **120**, the can **100** may have a negative polarity. Therefore, the positive electrode terminal **600** may be installed to be insulated from the can **100**.

[0056] The insulating member **500** may have a hollow circular plate shape and may be made of an insulating material. The insulating member **500** may be an insulating tape or an insulating plate. A lower portion of the positive electrode terminal **600** and the first current collector plate **300** may be in contact with and electrically connected through the hollow (e.g., hole) of the insulating member **500**. The insulating member **500** may be disposed to be in close contact with the inside of the top surface part **110**. In some embodiments, the insulating member **500** may have a size less than or equal to that of the top surface part **110**. In some embodiments, the insulating member **500** may be disposed to be in close contact with the top surface of the first current collector plate **300**. In some embodiments, the insulating member **500** may have a shape greater than or equal to that of the first current collector plate **300**.

[0057] Referring to FIGS. 1 and 2, the positive electrode terminal **600** may be a rivet terminal fixed to the top surface part **110** in a rivet manner at the inside or outside. The positive electrode terminal may be inserted into the top surface part. The positive electrode terminal **600** may be installed to be insulated from the top surface **110** of the can **100** by the first gasket **700** (e.g., made of an insulating material) that may be disposed therebetween. The positive electrode terminal **600** may include a

substantially cylindrical main body **610** and a head **620** provided by modifying a portion of the main body **610**. A lower end of the main body **610** may be disposed inside the top surface part **110**. The lower end of the main body **610** may be in contact with and electrically connected to the first current collector plate **300**. The head **620** may be deformed if riveted and disposed outside the top surface part **110**. The head **620** may be disposed substantially parallel to the top surface part **110** with the first gasket **700** therebetween.

[0058] Referring to FIGS. **1** and **2**, the first gasket **700** may insulate the positive electrode terminal **600** and the can **100** from each other. In some embodiments, the first gasket **700** may be constituted by an upper gasket **710**, an intermediate gasket **720**, and a lower gasket **730**. The upper gasket **710** may be disposed between the positive electrode terminal **600** and an outer surface of the top surface part **110**. The intermediate gasket **720** may be disposed between the terminal hole of the top surface part **110** and the main body **610** of the positive electrode terminal **600**. The lower gasket **730** may be disposed between the positive electrode terminal **600** and the insulating member **500**. A hollow (e.g., hole) may be defined in the upper gasket **710**, the intermediate gasket **720**, and the lower gasket **730** to allow the positive electrode terminal **600** to be inserted through the hollow (e.g., hole). In some embodiments, the upper gasket **710**, the intermediate gasket **720**, and the lower gasket **730** may be integrated with each other. The above-described cap assemblies (e.g., including cap plate **800** and second gasket **900**) may be disposed at a side opposite to the positive electrode terminal **600**.

[0059] Referring to FIGS. **2** to **7A**, each of the cap assemblies may include a cap plate **800** and/or a second gasket **900**.

[0060] Referring to FIGS. **2**, **5**, and **6**, the cap plate **800** may be a circular plate if viewed on a plane. The cap plate **800** may be made of the same material as the can **100** (e.g., steel, nickel-plated steel, steel alloy, aluminum, aluminum alloy, or cold sheet for deep drawing (SPCE), or a laminated film or plastic material that can constitute a pouch). The cap plate **800** may be coupled to the side part **120** by using the second gasket **900** as a medium.

[0061] A portion of the cap plate **800** may have a convex shape toward an opposite direction of the electrode assembly **200** (downward direction in FIG. **2**). In some embodiments, a portion of the cap plate **800** may have a concave shape in a direction toward the electrode assembly **200** (upward direction in FIG. **2**).

[0062] With reference to FIG. **2**, the convex portion in the downward direction may be defined as a first area **810**, and the concave portion in the upward direction may be defined as a second area **820**. The second area **820** may be integrated with the outside of the first area **810**. The second area may be disposed inside the first area. The first area **810** and the second area **820** may have a height difference therebetween. With reference to FIG. **2**, the first area **810** may protrude outward from the second area **820**. A notch **812** that functions as a vent may be provided on the first area **810** (e.g., the cap plate may include a notch disposed along a circumferential direction). Alternatively or additionally, a notch may be provided on the second area **820**. In some embodiments, a third area **830** may be provided to be integrated with the outside of the first area **810**. The third area **830** may be parallel to the second area **820**. That is, the third area may be arranged on the same straight line as the second area **820** with respect to FIG. **2**. As shown in FIG. **7A**, a coupling part **832** having a sawtooth shape in cross section (disposed) in a circumferential direction may be provided on the top surface of the third area **830**. The coupling part **832** may be manufactured through a knurling method (the coupling part will be described later). At least a portion of the third area **830** may be disposed between the beading part **122** and the crimping part **124** by using the second gasket **900** as a medium. Because the second gasket **900** (e.g., made of an insulating material) is disposed between the cap plate **800** and the side part **120**, the cap plate **800** may be insulated from the can **100**. In some embodiments, the cap plate **800** may become neutral (non-polar) having no negative or positive polarity. The third area **830** may be integrated with the second gasket **900** (this will be described later).

[0063] The notch **812** may be disposed adjacent to the third area **830**. The notch **812** may have a V-shaped cross section as a substantially circular ring-shaped groove. If viewed on the plane, the notch **812** may be provided as a circular groove in the cap plate **800**. The notch **812** may serve as a safety vent that is broken if a pressure inside the secondary battery **10** exceeds a certain level (e.g., a set pressure) to allow an internal gas to leak therefrom.

[0064] Referring to FIGS. **2** to **7A**, the second gasket **900** may have a shape that surrounds a portion of the cap plate **800**. The second gasket **900** may be integrated with the cap plate **800** by double injection (e.g., the gasket and cap plate may be provided through double injection). The second gasket **900** may be shaped to surround both upper and lower edges of the cap plate **800** (e.g., based on FIG. **3**). In more detail, an area surrounded by the second gasket **900** may be the third area **830** of the cap plate **800** (e.g., the gasket may be spaced apart from the notch). The second gasket **900** may be provided to surround the first area **810**, but it may be provided so as not to cover at least the notch **812**. In some embodiments, the second gasket **900** may have a ring shape (e.g., a substantially circular ring shape) as illustrated in FIG. **5** if viewed from above, and the second gasket may have a predetermined width. Referring to FIGS. **5** to **7A**, the second gasket **900** may include a first portion **910** disposed on a top surface of the third area **830** of the cap plate **800**, a second portion **920** disposed on a side surface of the third area **830**, a third portion **930** disposed on a bottom surface of the third area **830**, an extension portion **940** extending from the third portion **930** toward the electrode assembly **200**, and a fourth portion **950** disposed to be across the extension portion **940**. The first portion **910**, the second portion **920**, the third portion **930**, the extension portion **940**, and the fourth portion **950** may be integrated with each other.

[0065] Referring to FIG. **7A**, the first portion **910** may have a width corresponding to a horizontal width of the third area **830**. The above-described coupling part **832** may be disposed on the third area **830** (e.g., an area of the top surface that is in contact with the first portion). The coupling part **832** may have a sawtooth-shape in cross-section to increase in adhesion (e.g., enhance adhesion) with the second gasket **900** during the double injection. The sawtooth shape of the coupling part **832** may be uniformly disposed with the same size. Because the first portion **910** may be provided to be integrated with the cap plate **800**, the first portion **910** may also have a sawtooth-shape in cross-section corresponding to the shape of the coupling part **832** (e.g., the second gasket may be integrated with the cap plate via the coupling part). In some embodiments, the coupling part **832** may be provided in a shape other than the sawtooth shape. In some embodiments, the first portion **910** may also have a cross-sectional shape corresponding to the shape of the coupling part **832**.

[0066] The second portion **920** may correspond to a side surface of the second gasket **900** to connect the first portion **910** to the third portion **930**.

[0067] The third portion **930** may be connected to the second portion **920** and may be disposed on the bottom surface of the third area **830** of the cap plate **800**. A thickness of a portion of the third portion **930** may be less than that of the other portions (e.g., the first portion) to reduce the thickness.

[0068] The extension portion **940** may be a portion that protrudes from an end of the third portion **930** in a direction toward the electrode assembly **200**. The extension portion **940** may have a length that is not in contact with the second current collector (e.g., as shown in FIG. **2**). The extension portion **940** may allow the sealing force to be maintained even if the thickness of the third portion **930** is reduced.

[0069] The fourth portion **950** may have a bar shape having a predetermined width and may be a portion that connects one side of the extension portion **940** to the other side. The fourth portion **950** may be provided in a plurality. The reason why, in some embodiments, the fourth portion **950** does not cover the entire lower portion of the second gasket **900** is to reduce the thickness of the part at the bottom/lower portion of the cap plate **800**. In some embodiments, the fourth portion **950** may be disposed in an X or + (plus) shape if viewed from above. The exemplary cross-shaped part may be structured to prevent interference with the current collector.

[0070] FIG. 7B illustrates a partial cross-sectional view of the cap assembly of FIG. 7A, according to other embodiments.

[0071] Referring to FIG. 7B, in a cap plate **800a** according to other embodiments, a coupling part **832a** of a third area **830a** may have a different sawtooth-shaped density. In some embodiments, a partial area of the coupling part **832a** facing a center of the cap plate **800a** may have a tooth-shaped density greater than that of an area of the coupling part **832a** facing the outside of the cap plate **800a**. In some embodiments, a shape of the first portion **910a** of a second gasket **900a** (e.g., a gasket including second portion **920a**, third portion **930a**, and extension portion **940a**) may also have a shape corresponding to a cross-sectional shape of the coupling part **832a**.

[0072] In the embodiments of FIGS. 7A and 7B described above, if a thickness (height) of each of the third areas **830** and **830a** of the cap plates **800** and **800a** is about 1 T (1 mm), a height (thickness) of the sawtooth shape of each of the coupling parts **832** and **832a** may be about 0.3 T (0.3 mm). The height of each of the coupling parts **832** and **832a** may be changed and/or selected to be within about 30% of the thickness of each of the third areas **830** and **830a** of the cap plates **800** and **800a**.

[0073] As described above, the shape of the second gasket may be changed to reduce the thickness at the lower portion of the cap plate, thereby reducing a total thickness of the cap plate and the second gasket. Therefore, the size of the electrode assembly may increase as the thickness of the second gasket is reduced. In some embodiments, the capacity of the secondary battery may increase.

[0074] According to embodiments of the present disclosure, the cap plate and the gasket may be integrated with each other to reduce the costs and the assembly time.

[0075] In addition, the thickness of the gasket at the lower portion of the cap plate may be reduced to secure the space for improving the battery capacity while maintaining the sealing performance.

[0076] According to some embodiments, there is provided a method of manufacturing a secondary battery such as according to the above-mentioned embodiments, the method including: providing a can having one end opened; accommodating an electrode assembly in the can; coupling a cap plate to the one end of the can that is opened; and integrating a gasket with the cap plate, wherein the gasket is surrounding at least a portion of top and bottom surfaces of the cap plate.

[0077] The above-mentioned embodiments are merely exemplary embodiments, and thus, the present disclosure is not limited to the foregoing embodiments. Also, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the following claims.

Claims

1. A secondary battery comprising: a can having one end opened; an electrode assembly accommodated in the can; a cap plate coupled to the one end of the can; and a gasket surrounding at least a portion of a top surface and a bottom surface of the cap plate and integrated with the cap plate.
2. The secondary battery as claimed in claim 1, wherein the can has a cylindrical shape, and the cap plate has a circular plate shape.
3. The secondary battery as claimed in claim 2, wherein the gasket has a ring shape with a predetermined width and comprises a first portion disposed on the top surface of the cap plate, a third portion disposed on the bottom surface of the cap plate, and a second portion connecting the first portion to the third portion.
4. The secondary battery as claimed in claim 3, wherein the cap plate comprises a coupling part disposed on an area of the top surface that is in contact with the first portion.
5. The secondary battery as claimed in claim 4, wherein the coupling part has a sawtooth shape in cross-section.

- 6.** The secondary battery as claimed in claim 5, wherein the coupling part is disposed in a circumferential direction of the cap plate.
 - 7.** The secondary battery as claimed in claim 6, wherein the coupling part is manufactured through a knurling method.
 - 8.** The secondary battery as claimed in claim 7, wherein a height of the coupling part is within about 30% of a thickness of the cap plate.
 - 9.** The secondary battery as claimed in claim 7, wherein the gasket and the cap plate are provided through double injection.
 - 10.** The secondary battery as claimed in claim 9, wherein the gasket is made of an insulating material.
 - 11.** The secondary battery as claimed in claim 10, wherein, in the gasket, a thickness of the third portion is less than that of the first portion.
 - 12.** The secondary battery as claimed in claim 4, wherein the cap plate comprises a notch disposed along a circumferential direction.
 - 13.** The secondary battery as claimed in claim 12, wherein the cap plate comprises a first area, a second area disposed inside the first area and provided to be concave in a direction of the electrode assembly, and a third area disposed outside the first area and disposed in the same line as the second area.
 - 14.** The secondary battery as claimed in claim 13, wherein the notch is provided in the first area.
 - 15.** The secondary battery as claimed in claim 14, wherein the gasket is surrounding the third area and is spaced apart from the notch.
 - 16.** The secondary battery as claimed in claim 15, wherein the gasket further comprises an extension portion protruding from an end of the third portion toward the electrode assembly.
 - 17.** The secondary battery as claimed in claim 16, further comprising a first current collector plate electrically connected to a positive electrode plate of the electrode assembly; and a second current collector plate electrically connected to a negative electrode plate of the electrode assembly, wherein the extension portion has a length at which the extension portion is not in contact with the second current collector plate.
 - 18.** The secondary battery as claimed in claim 17, wherein the can comprises a circular top surface part and a side part extending downward from the top surface part, and the second current collector plate comprises a plurality of negative electrode leads electrically connected to the side part.
 - 19.** The secondary battery as claimed in claim 18, wherein the can further comprises: a beading part provided to be concave inward from the side part; and a crimping part spaced apart from the beading part and provided by bending an end of the side part inward, wherein a negative electrode lead of the plurality of negative electrode leads is inserted between the beading part and the second gasket.
 - 20.** The secondary battery as claimed in claim 18, further comprising: a positive electrode terminal inserted into the top surface part and electrically connected to the first current collector plate; and a gasket disposed between the top surface part and the positive electrode terminal and made of an insulating material.
-